

LAURA ANN BUSCH | Curriculum Vitae

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RESEARCH INTERESTS

Observational sub-mm astronomy, structure and evolution of the interstellar medium, the Galactic centre, (Galactic) star formation, protostars and protostellar outflows, astrochemistry, complex organic molecules.

CURRENT POSITION

POSTDOCTORAL POSITION

since 01/2024

Max Planck Institute for Extraterrestrial Physics (MPE) Garching

Supervised by Prof. Dr. Paola Caselli and Dr. Jaime Pineda.

EDUCATION

PH.D IN ASTRONOMY/ASTROPHYSICS

03/2020 - 12/2023

Max Planck Institute for Radio Astronomy (MPIfR) Bonn with University Bonn

Title: 'Formation and desorption of organic molecules: Resolving their emission in a prominent protostellar hot core and its outflow' written in the submm/mm department, supervised by Prof. Dr. Karl M. Menten and Dr. Arnaud Belloche.

MASTER OF SCIENCE IN ASTROPHYSICS

04/2017 - 02/2020

University Bonn

BACHELOR OF SCIENCE IN PHYSICS

10/2013 - 03/2017

University Bonn

EXPERIENCE (RESEARCH, TEACHING, AND OBSERVATION)

MSC THESIS

02/2019 - 02/2020

at MPIfR with University Bonn

Thesis: 'The peculiar ISM at the edge of the Central Molecular Zone' written in the submm/mm department at the MPIfR in Bonn, supervised by Prof. Dr. Karl M. Menten and Dr. Denise Riquelme.

INTERNSHIP

06/2018 - 01/2019

at MPIfR

Publication: 'The dynamically young outflow of the Class 0 protostar Cha-MMS1', published in A&A, supervised by Dr. Arnaud Belloche in the mm/submm department.

BSC THESIS

10/2016 - 03/2017

at MPIfR with University Bonn

Thesis: 'The obscuring torus of the elliptical galaxy NGC 1052' written in the VLBI department at the MPIfR in Bonn, supervised by Prof. Dr. Eduardo Ros.

TEACHING

at Argelander Institute for Astronomy Bonn

Tutorials and exam preparations as well as tutorial supervision for two lectures:
'Astro822: The physics of the interstellar medium', summer terms 2018 and 2021
'Astro121: Einführung in die Astronomie', winter terms 2017/18 and 2018/19

OBSERVATIONS TRAINING AT THE APEX 12 M AND THE IRAM 30M TELESCOPES
on site and remote

since 2019

OBSERVATIONS TRAINING AT THE EFFELSBERG 100 M
on site and remote

Winter 2023

MEMBERSHIPS

COLLABORATIVE RESEARCH CENTRE (CRC) 956

Group B3: Tracing Complex Molecules

03/2020 - 12/2022

INTERNATIONAL MAX PLANCK RESEARCH SCHOOL FOR ASTRONOMY AND ASTROPHYSICS

03/2020 - 12/2023

BONN-COLOGNE GRADUATE SCHOOL FOR PHYSICS AND ASTRONOMY

04/2017 - 12/2023

PRESENTATIONS

Contributed talk, 839. WE-Heraeus-Seminar: Charting the cosmos: From cosmic stellar nurseries to evolved stars using high powered telescopes, 08/2025, Ingelheim, Germany. Title: 'From clouds to stars: What $^{12}\text{C}/^{13}\text{C}$ ratios reveal about molecular inheritance'.

Contributed talk, Fractionation of isotopes in space II – from the Solar System to galaxies, 11/2024, Florence, Italy. Title: 'Do we observe ^{13}C enrichment towards young low-mass protostars?'.

Contributed talk, The European Astronomical Society Annual Meeting, 07/2024, Padova, Italy. Title: 'Shocking Sgr B2 (N1) with its own outflow - A new perspective on O/N segregation in molecules'.

Contributed talk, Connecting the hot core and hot corino phenomena II, 11/2022, Bordeaux, France. Title: 'Resolving emission of complex organic molecules in the hot core Sgr B2(N1) and its outflow with ReMoCA'.

Contributed talk, From Clouds to Planets II: The Astrochemical Link, 10/2022, Berlin, Germany. Title: 'Exposing the COMfort zone of Sgr B2 (N1) by resolving the desorption of Complex Organic Molecules in the hot core'.

Solicited seminar, Journal club at the Max Planck Institute for Extraterrestrial Physics, virtual, on [A&A](#), [665](#), [A96](#), 09/2022.

Contributed talk, The European Astronomical Society Annual Meeting, 06/2022, Valencia, Spain. Title: 'Exposing the COMfort zone of Sgr B2 (N1) by resolving the desorption of Complex Organic Molecules in the hot core'.

Contributed talk, Meeting of ALMA Young Astronomers, 03/2022, virtually. Title: 'Resolving the COMfort zone of Sgr B2 (N1) with ReMoCA'.

Contributed talk, Science with the Atacama Pathfinder EXperiment (APEX), 02/2020, Ringberg, Germany. Title: 'High- and low-velocity gas at the edge of the Central Molecular Zone'.

WORKSHOPS & SUMMER SCHOOLS

ASTROCHEMISTRY IN THE BROADBAND ERA: NGVLA & ALMA WSU

Portland, USA

10/2025

INTERSTELLAR DISPARITY: N VS O-BEARING MOLECULES

Lorentz-Centre, Leiden, Netherlands

08/2025

Poster presented entitled: 'New clues on O/N segregation from Sgr B2 (N1) and its outflow'

11TH IRAM INTERFEROMETRY SCHOOL

Grenoble, France

11/2022

PRINCIPLES AND TOOLS OF STATISTICS, DATA ANALYSIS AND MACHINE LEARNING:

I.Statistics, MPIfR Bonn, Gunther Witzel

04/2022

INTERNATIONAL SUMMER SCHOOL ON THE INTERSTELLAR MEDIUM

of Galaxies, from the Epoch of Reionisation to the Milky Way (virtual)

07/2021

ACCEPTED OBSERVING PROPOSAL

100 m Effelsberg Telescope

'Hot, hotter, gas at the edge of the CMZ?' (16 hours),

Observations carried out in Jan/Feb 2023

12 m APEX Telescope

'Shocking! Complex organic molecules at the edge of the CMZ' (36 hours),

Observations carried out in spring 2023 (57/36 hours)

SKILLS

OS	Mac OS, Linux, Windows
PROGRAMMING LANGUAGES	Python (Numpy, Matplotlib, SciPy, AstroPy, SpectralCube), LaTeX
SOFTWARES	Gildas (Class, Weeds, Greg, Mapping), Madcuba, Radex
LANGUAGES	Native: German Fluent: English Good knowledge: French Beginner: Spanish

EXTRA CURRICULAR ACTIVITIES

PUBLIC TALKS AT BAD MÜNSTEREIFEL

Talk series in German for the interested public audience

04/2023

MONTAGSVORTRÄGE IN DER STERNWARTE

Talk series in German for the interested public audience

03/2023

LOC IMPRS STUDENT SEMINAR

Organising and hosting the student seminar of IMPRS in Bonn, in virtual only and hybrid mode

07/2021 & 01/2023

HIGHLIGHTS DER PHYSIK

In collaboration with MPIfR Bonn

Representing the research departments of MPIfR at the exposition during this event, which was initiated to promote research in physics and astronomy amongst the public.

09/2019

GIRL'S DAY

In collaboration with MPIfR Bonn

Conducting workshops to promote science amongst female school students.

2017, 2018, 2023